

Restock the Pantry Shelves SOP (1x Restock, in situ)

One episode restocks one pantry shelf, in place, and flattens the box the stock came in.

The base is **passive**. It has no drive of its own. It is pushed by hand to the pantry unit and locked there. Nothing is carried away to a table. Everything the episode touches is already at the pantry when recording starts: the older stock standing on the working shelf, and the six new items standing in the delivery box on the stand in front of the base.

The pantry is worked **as found**. The working shelf carries three zones, each marked by a **zone label** on the shelf front edge: **CARTONS** at the left end, **JARS** in the middle, **CANS** at the right end. Each zone holds one older item at its back. The delivery box holds two new items of each kind. The episode ends with all six new items shelved in their zones, every item label facing the front, every older item standing in front of the new stock behind it, and the empty delivery box lying flat on the stand.

This is an in-situ task. Three rules follow from that.

First, there is a shelf directly above the working shelf. So **every approach to the shelf is from the front, straight in, level**. No gripper comes down onto the working shelf from above.

Second, **the pantry unit is never leaned on and never pushed**. No gripper, wrist, or forearm rests on a shelf edge, a shelf board, or a zone label. No push is ever hard enough to move the unit.

Third, **older stock is slid, never lifted**. It is drawn forward along the shelf board. Nothing that starts on the working shelf comes off it.

Every zone is loaded the same way: rotate first, then fill. The older item in that zone is drawn forward to the front, and the new items go in behind it. New stock never goes in front of older stock, anywhere.

Every item is faced. An item stands on the shelf with its item label facing the front. A new item is turned to face the front while the gripper still holds it on the shelf, before the gripper opens.

The order never changes. Load the cartons zone. Load the jars zone. Load the cans zone. Flatten the delivery box. The box is flattened last, so it is never flattened with stock still in it and nothing is ever carried over a flattened box.

The two arms never cross. The **left gripper** works the cartons zone at the left end of the shelf and the left column of the delivery box. The **right gripper** works the jars zone, the cans zone, the middle and right columns of the delivery box, and the whole of the flatten. The left gripper always stays left of the right gripper.

Setup

Complete the base positioning and both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Base positioning

The base is **passive**. It has no drive of its own. It is pushed into place by hand. It is never steered, nudged, or repositioned once recording starts. It is parked once, before recording, and does not move again until the episode is over.

1. Push the base by hand up to the pantry unit. Stop it **square to the front of the unit**, so the front edge of the working shelf runs straight across the frame and neither end of it sits nearer than the other.
2. Stop it **centered on the working shelf**, so the left end and the right end of the shelf are the same distance out from the middle of the base.
3. Stop it **close enough** that both grippers reach the back of the working shelf straight in through the shelf opening without an arm extending. Stop it **far enough** that no arm, wrist, or part of the base touches the unit while both arms work.
4. Check the **height band**. With the base parked, both grippers enter the shelf opening level and reach the shelf board from front to back, carrying an item, without a wrist or forearm touching the shelf above. Both grippers also reach straight down into the delivery box on the stand.
5. Check the **left lane**. The **left gripper** reaches the cartons zone at the left end of the shelf and the left column of the delivery box, all without extending and without reaching right of the middle of the base.
6. Check the **right lane**. The **right gripper** reaches the jars zone, the cans zone, the middle and right columns of the delivery box, and both corners of the box it pushes in Step 4, all without extending and without reaching left of the middle of the base.
7. Check the **stand**. The delivery box stands on it below and clear of the shelf front edge, and there is clear stand top around the box for the flattened box to lie on.
8. Lock or brake the base. Push it firmly once by hand. It must not roll, creep, or turn.
9. If any of lines 1 to 7 fails, push the base to a new park by hand and start again at line 1. Do not work a shelf the arms cannot reach comfortably.

The base stays locked and still for the whole episode. Nothing moves it. No arm leans on the unit hard enough to shift it. It is never repositioned mid-task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the shelf opening. Its frame includes the whole working shelf, the three zone labels on the shelf front edge, the shelf above, and the delivery box on its stand.
3. The camera reads the working shelf from the front. You can see which zone an item stands in, whether it stands upright, whether its item label faces the front, and whether anything sticks out past the shelf front edge.
4. The camera reads the three zone labels. You can read CARTONS, JARS, and CANS in frame.
5. The camera reads the delivery box from the front and above. You can see which column an item stands in and whether the box is empty, and you can see the box lying flat at the end.
6. Both arms are at home with grippers open.
7. The **left arm** reaches the left column of the delivery box and the cartons zone without extending to a joint limit. It never needs to reach the jars zone, the cans zone, or the middle or right column of the box.
8. The **right arm** reaches the middle and right columns of the delivery box, the jars zone, the cans zone, and both corners of the box used in the flatten, without extending to a joint limit. It never needs to reach the cartons zone or the left column of the box.
9. Each gripper enters and leaves the shelf opening **from the front and level**. Each gripper lifts an item out of the delivery box straight up. Neither gripper comes down onto the working shelf from above, and neither wrist nor forearm touches the shelf above on the way in or out.
10. The two arms do not touch each other. The left gripper stays left of the right gripper throughout.
11. If a place cannot be reached, re-park the base by the Base positioning steps until lines 7 to 10 hold.

Materials checklist

1. The **pantry unit** stands where it lives. It is not moved, not leaned on, and not pushed at any point.
2. The **working shelf** is one shelf of the unit with a **shelf directly above it**. So the only way in is the front-facing shelf opening, never the top.
3. There is clear space above the working shelf, so the tallest item stands on it without touching the shelf above, and a gripper carrying an item enters level over a standing item without touching that shelf.
4. The working shelf is dry, level, and bare apart from the older stock. Nothing on it is taped or marked other than the three zone labels.

5. Three **zone labels** are fixed to the **shelf front edge**, one under each zone, reading **CARTONS** at the left end, **JARS** in the middle, and **CANS** at the right end. Each label is stuck down firmly, reads from the front, and is never touched by a gripper.
6. The working shelf carries **three older items**, standing upright, one at the back of each zone, none touching another: **one older carton** in the cartons zone, **one older jar** in the jars zone, **one older can** in the cans zone. Each older item stands with its item label facing the front.
7. The three kinds are visibly different from the front. A carton, a jar, and a can are never a close call.
8. Each zone is deep enough front to back for three items of its kind to stand in one row with clear space between them, and no wider than one item stands comfortably. So each zone is one row and there is no choice of lane.
9. The **stand** stands on the floor directly in front of the base, below and clear of the shelf front edge, at a height where both grippers lift an item straight up out of the box without extending. It does not slide when an item is lifted or when the box is pushed. Its top is large enough for the flattened box to lie on it flat.
10. The **delivery box** stands on the stand, centered left to right on the base, its open top facing up. It is a plain cardboard box with **no top flaps**. Its **bottom flaps are folded in loosely and not taped and not tucked into each other**, so the box collapses flat when two opposite corners are pushed toward each other. **Unvalidated**.
11. The box is tall enough that an item standing in it does not show above the rim by more than the height the gripper needs to close on its body. It is not so tall that the gripper cannot reach the box floor.
12. The box holds the **six new items** standing upright in **three columns of two**, front and back, none on top of another and none touching:
 - **left column**: two cartons;
 - **middle column**: two jars;
 - **right column**: two cans.
13. Each new item matches one older item by kind, and its column matches its zone, so where it goes is never a judgment call: cartons to the cartons zone, jars to the jars zone, cans to the cans zone.
14. Every item in the box's left column belongs to the zone the **left gripper** works. Every item in the middle and right columns belongs to a zone the **right gripper** works. No item ever crosses a column.
15. Each new item stands in the box with its item label turned **no more than a quarter turn** away from the front, to either side. No item label faces the back.
16. Every item, older and new, stands on its own base unaided, is closed and dry, carries one printed item label on its side, and is held by one gripper around its body without its contents

spilling or its lid or cap coming loose.

17. The floor and the space between the stand and the shelf opening are clear, so an arm carrying an item passes over nothing.

Workspace layout

Nothing is marked or taped out except the three zone labels. You judge every place below by eye against the unit itself: the shelf opening, the shelf front edge, the zone labels, and the stock already standing in place.

- **Pantry unit:** the shelf unit the base is parked at. It is never moved, never leaned on, and never pushed.
- **Shelf opening:** the open front of the working shelf. Both grippers go in and come out through it, level and from the front.
- **Working shelf:** the one shelf worked. All shelf work happens on it.
- **Shelf above:** the shelf directly above the working shelf. It is what makes every shelf approach a front approach. Nothing ever touches it.
- **Shelf front edge:** the front edge of the working shelf board, carrying the three zone labels. Nothing may stick out past it.
- **Cartons zone:** the stretch of the working shelf at its **left end**, under the CARTONS label, where the older carton stands. **Left gripper only.**
- **Jars zone:** the stretch in the **middle**, under the JARS label, where the older jar stands. **Right gripper only.**
- **Cans zone:** the stretch at the **right end**, under the CANS label, where the older can stands. **Right gripper only.**
- **Stand:** the stand in front of the base holding the delivery box. The flattened box lies on it at the end.
- **Delivery box:** the box on the stand. Its **left column** holds the two cartons. Its **middle column** holds the two jars. Its **right column** holds the two cans. Empty and flat at episode end.

Arm lanes

The two arms never cross. This holds for the whole episode.

- The **left gripper works the left lane:** the left column of the delivery box and the cartons zone.
- The **right gripper works the right lane:** the middle and right columns of the delivery box, the jars zone, the cans zone, and the flatten of the delivery box.
- The **left gripper always stays left of the right gripper.** Neither arm reaches over, under, around, or past the other. Neither arm reaches across the front of the other arm's body.

- Nothing is handed between grippers, and no item ever crosses a column. Only one item is moved at a time.
- In the flatten, the **left gripper** holds one corner of the box still and the **right gripper** pushes the opposite corner. Each stays on its own side of the box's middle.

Arm assignments

- **Left gripper.** Takes the two cartons from the box's left column. Rotates and loads the cartons zone. Holds the back-left corner of the empty box still while it is flattened in Step 4.
- **Right gripper.** Takes the two jars from the middle column and the two cans from the right column. Rotates and loads the jars zone and the cans zone. Pushes the front-right corner of the empty box to collapse it, and lays the flat box down on the stand, in Step 4.

Vocabulary

- **Older stock:** the three items already standing on the working shelf when recording starts. Older stock never comes off the shelf.
- **New stock:** the six items standing in the delivery box when recording starts. All six go onto the shelf during the episode.
- **Zone:** one of the three stretches of the working shelf, each marked by a zone label on the shelf front edge. A new item goes to the zone whose label names its kind.
- **Zone label:** the printed label stuck to the shelf front edge under a zone, reading CARTONS, JARS, or CANS. It is read, never touched.
- **Item label:** the printed label on the side of a carton, jar, or can. It is what must face the front.
- **Faced:** the item stands with its item label facing the front, so the label reads square on from the shelf opening. An item whose label faces a side wall or the back is not faced.
- **Rotate forward:** draw the older item in that zone toward the front of the zone, so it stands in front of everything that goes in after it. It is done before any new item goes into that zone.
- **Behind:** further from the shelf front edge, nearer the back of the shelf.
- **Front approach:** the gripper enters and leaves the shelf opening level and from the front. It never comes down onto the working shelf from above. Every action on the working shelf is a front approach.
- **Body grip:** the gripper closes around the side of the item, below its lid, cap, or top seam, where the item is solid. Never on a lid, cap, or carton flap.
- **Column:** one of the three front-to-back pairs of items in the delivery box. Left, middle, right.
- **Front item** and **back item:** in a column, the item nearer the front edge and the item nearer the shelf. The front item is always taken first.
- **Stands on its own:** after the gripper moves clear, the item stays upright for 2 seconds and does not rock, lean, slide, or fall.

- **Upright and square:** the item stands on its own base, faced, not leaning, not lying on its side, and not turned to face a side wall.
- **Clear space:** two items in the same zone do not touch each other, and the gap between them is visible from the front.
- **Overhang the front edge:** any part of an item sticks out past the shelf front edge into the shelf opening.
- **Collapsed:** the box's four walls have folded flat against each other, so the box is a flat two-ply panel with no wall standing up and no space inside. **Unvalidated.**
- **Lying flat:** the collapsed box lies on the stand on one of its faces, with nothing under it and no edge lifted.

Steps

Step 1: Load the cartons zone

Goal: the cartons zone holds the older carton at its front and both new cartons behind it, all upright and square, faced, with clear space between them and nothing overhanging the shelf front edge.

The cartons zone is worked by the **left gripper** alone. The **right arm stays at home** for the whole of this step.

1.1 Rotate the older carton forward

- With the **left gripper** open, enter the shelf opening from the front, level, and close on the older carton in a body grip.
- Draw the carton forward along the shelf board until it stands just inside the shelf front edge. Keep it on the board the whole way. Do not lift it. Do not tip it. Do not take it off the shelf.
- Open the **left gripper** and draw it straight out of the shelf opening.

Check: the older carton stands upright and square just inside the shelf front edge, faced, nothing sticks out past that edge, and there is clear shelf behind it. If it leans, stand it up with the **left gripper**. If it stands out past the edge, press it back with the **left gripper** closed. If its label has turned, close on it again in a body grip, turn it until it is faced, and open.

1.2 Stand the first new carton at the back

- With the **left gripper**, close on the **front item** of the box's left column in a body grip.
- Lift it straight up out of the box. Never dig under another item. Never lift two items together.
- Carry it level to the shelf opening and take it in from the front, high enough to clear the older carton, no higher. Do not carry it across the middle of the base.
- Stand it down on the shelf board at the **back of the cartons zone**, upright, with clear space between it and the older carton.
- Keep the grip. Turn the carton until it is faced. Only then open the **left gripper**.

- Draw the **left gripper** straight out of the shelf opening.

Check: the new carton stands on its own at the back of the zone, faced, not touching the older carton. If it leans or touches, close on it again with the **left gripper**, stand it down again, and face it.

1.3 Stand the second new carton between

- With the **left gripper**, close on the **back item** of the box's left column in a body grip and lift it straight up out of the box.
- Carry it level to the shelf opening and take it in from the front, high enough to clear the older carton.
- Stand it down **between the older carton and the first new carton**, upright, with clear space on both sides.
- Keep the grip. Turn the carton until it is faced. Only then open the **left gripper**.
- Draw the **left gripper** straight out of the shelf opening.

Check: three cartons stand in one row in the cartons zone, older at the front and both new behind it, all upright and square, all faced, clear space between each pair, and nothing overhanging the shelf front edge. The box's left column is empty. If a carton leans, touches its neighbour, or is not faced, close on it again with the **left gripper**, set it right, and open.

Expected state: cartons zone full, jars zone and cans zone still holding their older item only, four new items left in the box's middle and right columns.

Step 2: Load the jars zone

Goal: the jars zone holds the older jar at its front and both new jars behind it, all upright and square, faced, with clear space between them and nothing overhanging the shelf front edge.

The jars zone is worked by the **right gripper** alone. The **left arm stays at home** from here to the end of Step 3.

- Run the same motions as Step 1 with the **right gripper**, using the jars zone and the **middle column** of the box.
- Rotate the older jar forward to just inside the shelf front edge in a body grip, without lifting it.
- Take the **front item** of the middle column first and stand it at the back of the jars zone. Take the **back item** second and stand it between the older jar and the first new jar.
- Face each new jar while the grip is still closed, then open.

Check: three jars stand in one row in the jars zone, older at the front and both new behind it, all upright and square, all faced, clear space between each pair, and nothing overhanging the shelf front edge. The box's middle column is empty. Set any fault right with the **right gripper** before moving on.

Expected state: cartons zone and jars zone full, cans zone still holding its older can only, two new cans left in the box's right column.

Step 3: Load the cans zone

Goal: the cans zone holds the older can at its front and both new cans behind it, all upright and square, faced, with clear space between them and nothing overhanging the shelf front edge.

- Run the same motions as Step 1 with the **right gripper**, using the cans zone and the **right column** of the box.
- Rotate the older can forward to just inside the shelf front edge in a body grip, without lifting it.
- Take the **front item** of the right column first and stand it at the back of the cans zone. Take the **back item** second and stand it between the older can and the first new can.
- Face each new can while the grip is still closed, then open.

Check: three cans stand in one row in the cans zone, older at the front and both new behind it, all upright and square, all faced, clear space between each pair, and nothing overhanging the shelf front edge. The delivery box is empty. Set any fault right with the **right gripper** before moving on.

Expected state: all six new items shelved, nine items on the working shelf in three rows of three, the delivery box standing empty on the stand.

Step 4: Flatten the empty delivery box

Goal: the delivery box is collapsed and lying flat on the stand.

The box is collapsed by pushing two opposite corners toward each other. The **left gripper** holds the back-left corner still. The **right gripper** pushes the front-right corner. The box stays on the stand the whole time. It is never lifted off the stand and never carried.

4.1 Check the box is empty

- Read the box from the front. All three columns are empty. Nothing lies on the box floor.
- Read the shelf. Nine items stand in three rows, nothing overhangs the shelf front edge, and both grippers are out of the shelf opening and empty.

Check: if any item is still in the box, go back to the step for its zone and shelve it before going on.

4.2 Collapse the box

- With the **left gripper** closed, set it against the outside of the box's **back-left corner**, low down near the box floor. Hold it there. It does not push. It only stops the box sliding.
- With the **right gripper** closed, set it against the outside of the box's **front-right corner**, low down near the box floor.

- Push the front-right corner steadily **toward the back-left corner** in one smooth move, until the walls fold flat against each other and the box is collapsed. Do not push fast. Do not lift either corner.
- Stop pushing as soon as the box is collapsed. Do not press the flat box into the stand.
- Draw the **left gripper** straight back off the box and keep it clear until the episode ends.

Check: the box is collapsed. No wall stands up, and no space shows inside. The base and the stand have not moved. If a wall stays part open, set the **right gripper** on the front-right corner again and push once more. If the bottom flaps hold the box open, do not force it. Leave it standing and report it.

4.3 Lay the box flat

- With the **right gripper** closed, set it against the top edge of the collapsed box on its right end.
- Tip the box over onto its face, away from the front edge, so it lies flat on the stand.
- Press its top face once, flat, so it lies with no edge lifted. Press only as hard as the box needs.
- Draw the **right gripper** straight back off the box.

Check: the box is lying flat on the stand, fully on the stand top, with nothing under it and no edge lifted. If an edge is lifted, press it once more with the **right gripper**. If the box lies part off the stand, push it onto the stand with the **right gripper** closed.

Expected state: nine items on the working shelf, three rows of three, older at the front of each zone, all faced. The box flat on the stand. Base still locked in its park.

Step 5: End the episode

- Confirm the shelf. Each zone holds three items of its own kind in one row, older at the front, all upright and square, all faced, with clear space between them.
- Confirm the front edge. Nothing overhangs it and all three zone labels are in place.
- Confirm the box. It is collapsed and lying flat on the stand.
- Confirm nothing has been left standing on the stand or dropped on the floor.
- Confirm the base has not moved and the unit has not shifted.
- Correct any failed check before ending.
- Return both arms home with grippers open. Then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the six new items off the working shelf: both new cartons, both new jars, both new cans.

2. Slide each older carton, jar, and can back to the back of its own zone, upright, faced, with the three zones clear of one another.
3. Pick up the flattened box, open it out, and fold its bottom flaps in loosely again, not taped and not tucked into each other. Stand it on the stand, open top up, centered left to right on the base.
4. Stand the six new items in the box in three columns of two: cartons left, jars middle, cans right, none touching. Turn each one so its item label is no more than a quarter turn from the front.
5. Check no item stands on the wrong side of a column.
6. Wipe up any spill on the shelf or the stand, and dry it.
7. Replace an item that leaks, that no longer stands on its own base, or whose lid or cap has come loose. Replace the box when it no longer collapses under a correct push or no longer stands open on its own.
8. Check all three zone labels are still stuck down firmly and read from the front. Stick a lifted one back down.
9. Check the stand is still steady and does not slide when an item is lifted out of the box or the box is pushed.
10. Check the base is still locked and parked square. Then run the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

- **Visible cue:** the pantry unit shifts in frame, the shelf opening changes angle or size in frame, or the base rolls, creeps, or turns at any point after recording starts.
- **SOP rule broken:** Steps 1 to 5. The base is parked and locked before recording and stays still for the whole episode.

- **Coaching note:** park it, lock it, push-test it, then start recording. A base that drifts is a lost episode.

Violation: Arms crossed

- **Visible cue:** the **left gripper** touches the jars zone, the cans zone, or the middle or right column of the box. The **right gripper** touches the cartons zone or the left column of the box. The left gripper goes to the right of the right gripper. One arm reaches over, under, around, or past the other, or across the front of the other arm's body.
- **SOP rule broken:** Steps 1 to 4. The left gripper works the left lane, the right gripper works the right lane, and the left gripper always stays left of the right gripper.
- **Coaching note:** the column tells you whose item it is. If you have to reach past the other arm, the wrong arm is doing the job.

Violation: Approached the shelf from above or touched the shelf above

- **Visible cue:** a gripper comes down onto the working shelf from above instead of entering level from the front. Or a wrist or forearm knocks, scrapes, or rests on the shelf above or the shelf front edge on the way in or out.
- **SOP rule broken:** Steps 1 to 3. Every action on the working shelf is a front approach, level, in and out through the shelf opening.
- **Coaching note:** straight in from the front, straight out the same way. There is no way in from the top.

Violation: Leaned on or pushed the unit

- **Visible cue:** a gripper, wrist, or forearm rests on a shelf board, a shelf edge, or the side of the unit. The unit rocks or shifts. Or a press on an item moves the whole unit.
- **SOP rule broken:** Steps 1 to 3. The pantry unit carries no weight and is never pushed out of place.
- **Coaching note:** the arm holds itself up. Press only as hard as the item needs.

Violation: Wrong loading order

- **Visible cue:** a zone is loaded before the zone to its left is done. The box is flattened before all six items are shelved. The back item of a column is taken before the front item. Or a zone is skipped and come back to.
- **SOP rule broken:** Steps 1 to 4. Load the cartons zone, then the jars zone, then the cans zone, front item before back item, and flatten the box last.
- **Coaching note:** left to right, front before back, box last. The order is the same every episode.

Violation: Older stock not rotated forward

- **Visible cue:** a new item goes into a zone while the older item there is still at the back. Or the older item is only part way forward and is left there.
- **SOP rule broken:** Steps 1.1, 2, and 3. Draw the older item forward to just inside the shelf front edge before any new item goes in behind it.
- **Coaching note:** rotate first, then fill. Every zone gets the old one moved before the new one arrives.

Violation: New stock put in front of older stock

- **Visible cue:** a new item ends up nearer the shelf front edge than the older item of that zone.
- **SOP rule broken:** Steps 1.2, 1.3, 2, and 3. The new items are stood down behind the older item every time.
- **Coaching note:** the older item is always the one you see first from the front. If a new one is in front, it is wrong.

Violation: Older stock lifted or taken off the shelf

- **Visible cue:** an older item is lifted off the shelf board, carried in the air, tipped over, or set down anywhere other than where it was drawn to.
- **SOP rule broken:** Steps 1.1, 2, and 3. Older stock is slid along the shelf board and never leaves the working shelf.
- **Coaching note:** slide it, do not lift it. Nothing that starts on the shelf comes off.

Violation: Item put in the wrong zone

- **Visible cue:** a gripper stands a carton under the JARS or CANS label, a jar under the CARTONS or CANS label, or a can under the CARTONS or JARS label. Or a new item is stood anywhere other than in a row with the older item of its own kind.
- **SOP rule broken:** Steps 1.2, 1.3, 2, and 3. Each new item goes to the zone whose label names its kind.
- **Coaching note:** read the zone label before the arm goes in. The label on the shelf edge decides, not the nearest gap.

Violation: Item not faced

- **Visible cue:** a gripper opens on a new item whose item label faces a side wall or the back. An older item is left with its label turned after the rotate. Or a gripper opens before the turn is made and then the item is left as it is.
- **SOP rule broken:** Steps 1.1, 1.2, 1.3, 2, and 3. Every item stands faced. A new item is turned to face the front while the grip is still closed, before the gripper opens.
- **Coaching note:** turn it, then let go. A label that faces the wall is a shelf nobody can read.

Violation: Item not stood upright and square

- **Visible cue:** an item on the shelf leans, lies on its side, touches its neighbour, or rocks after the gripper moves clear.
- **SOP rule broken:** Steps 1.2, 1.3, 2, and 3. Stand every item on its own base with clear space between each pair in its zone.
- **Coaching note:** stand it down, watch it settle, then let go.

Violation: Item left overhanging the shelf front edge

- **Visible cue:** an item on the working shelf sticks out past the shelf front edge into the shelf opening when the gripper moves on. Or an item stands on a zone label.
- **SOP rule broken:** Steps 1.1, 2, and 3. Every item stands inside the shelf front edge, with nothing sticking out past it.
- **Coaching note:** just inside the edge, not on it. Press it back before you move on.

Violation: Item taken from the box wrong

- **Visible cue:** a gripper digs under an item, lifts two items together, drags an item up the box wall, takes an item from another column, or knocks or tips the box or the stand while lifting.
- **SOP rule broken:** Steps 1.2, 1.3, 2, and 3. Take one item at a time, from your own column, and lift it straight up out of the box.
- **Coaching note:** one item, straight up, your own column, and the box stays where it stands.

Violation: Item gripped by its lid, cap, or flap

- **Visible cue:** a gripper closes on a jar lid, a can top seam, or a carton flap instead of around the item's body. Or a lid or cap comes loose or lifts off as the item is carried.
- **SOP rule broken:** Steps 1.1, 1.2, 1.3, 2, and 3. Close around the item's body in a body grip, not on its lid, cap, or flap.
- **Coaching note:** grip the body low, where the item is solid. A lid is not a handle.

Violation: Item carried wrong

- **Visible cue:** a gripper carries an item across the middle of the base, over the other arm's lane, over the stand's far side, or higher than it needs to clear the older item. Or it swings the item in on an arc instead of taking it in level from the front.
- **SOP rule broken:** Steps 1.2, 1.3, 2, and 3. Carry each item level to the shelf opening and take it in from the front, high enough to clear the older item and no higher.
- **Coaching note:** straight up, straight across, straight in. The shelf above is closer than it looks.

Violation: Box flattened with stock still in it

- **Visible cue:** the **right gripper** pushes a corner of the box while any item is still standing in it. Or Step 4 begins with an item still standing in a column.

- **SOP rule broken:** Step 4.1. Read the box and the shelf, and shelve any item still in the box, before the box is touched.
- **Coaching note:** empty first, then flatten. Count three rows of three on the shelf before the arm goes to the box.

Violation: Box collapsed wrong

- **Visible cue:** the **right gripper** pushes a corner other than the front-right one, or pushes it in a direction other than toward the back-left corner. The **left gripper** pushes instead of holding, or holds a corner other than the back-left one. A corner is lifted, the box is lifted off the stand, the push is fast or jerky, the box is torn, or the flat box is pressed into the stand.
- **SOP rule broken:** Step 4.2. The left gripper holds the back-left corner still and the right gripper pushes the front-right corner toward it, steadily, until the box is collapsed, with the box never leaving the stand.
- **Coaching note:** one corner held, the opposite corner pushed, one smooth move. The box folds itself if you let it.

Violation: Box not laid flat

- **Visible cue:** the collapsed box is left standing on edge, lies part off the stand, lies with an edge lifted, or is tipped toward the front edge instead of away from it. Or the box is laid down by the **left gripper**.
- **SOP rule broken:** Step 4.3. The right gripper tips the collapsed box away from the front edge onto its face and presses it once so it lies flat on the stand.
- **Coaching note:** tip it away, press it once, look at every edge. Flat means nothing lifted.

Violation: Box or stand knocked out of place

- **Visible cue:** the box slides, turns, or tips on the stand during Steps 1 to 3. The stand is pushed, turned, or rocked by an arm. Or the box is pushed off the stand during the flatten.
- **SOP rule broken:** Steps 1 to 4. The box and the stand stay where they stand until the flatten, and the box stays on the stand through the flatten.
- **Coaching note:** lift straight up so nothing drags. In the flatten, the held corner is what stops the box walking.

Violation: Zone label touched or disturbed

- **Visible cue:** a gripper, wrist, or an item touches, rubs, lifts, or peels a zone label. A label ends the episode lifted, moved, or covered by an item.
- **SOP rule broken:** Steps 1 to 3. The zone labels are read and never touched, and every item stands behind the shelf front edge, clear of them.
- **Coaching note:** the label is the map. Read it from the front and keep the gripper above the edge.

Violation: New stock left in the box

- **Visible cue:** the episode moves on to Step 5 with an item still standing in the box, or an item is found under the flattened box.
- **SOP rule broken:** Steps 1 to 3. All six new items go onto the working shelf: two cartons, two jars, two cans.
- **Coaching note:** read all three columns before the box is touched. It is empty when the shelving is done.

Violation: Required check not followed

- **Visible cue:** a check named in a step is skipped. Or a check is made and the fault it finds is left uncorrected: a leaning item, an item not faced, a wrong zone, an item over the front edge, a wall of the box still standing, or an edge of the flat box lifted.
- **SOP rule broken:** Steps 1.1 to 4.3. Run each check and correct what it finds by the retry written in that step.
- **Coaching note:** a check is not done until what it found has been put right.

Violation: Dropped, toppled, or knocked over

- **Visible cue:** an item is dropped on the stand, into the box, or on the floor. An item already standing is toppled by an arm or by another item. Or an item on the shelf falls off the shelf front edge.
- **SOP rule broken:** Steps 1 to 4. Nothing is dropped, toppled, or knocked out of its place, and every gripper leaves the shelf opening straight out to the front.
- **Coaching note:** check the path and the landing place before the arm moves, and come out the way you went in.

Violation: Wrong arm used

- **Visible cue:** the **left gripper** works the jars zone or the cans zone, takes an item from the middle or right column, pushes the box corner, or lays the box flat. The **right gripper** works the cartons zone, takes an item from the left column, or holds the back-left corner. Or an item is handed from one gripper to the other.
- **SOP rule broken:** Steps 1 to 4. The **left gripper** works the cartons zone, the left column, and the held corner. The **right gripper** works the jars zone, the cans zone, the middle and right columns, the pushed corner, and the lay-flat. Nothing changes hands.
- **Coaching note:** left arm left, right arm right, and nothing changes hands.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with an item still in the box, a zone short of three items, an item leaning, not faced, or in the wrong zone, the box standing or not flat, the stand knocked out of place, an arm short of home, or a gripper not fully open.

- **SOP rule broken:** Step 5. Confirm the shelf, the front edge, the box, the floor, and the base. Then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot read a zone label, a zone, or the box**, so where an item went, whether it is faced, or whether the box is empty cannot be judged.
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Base fault:** a brake or lock that will not hold, or a base that rolls, creeps, or turns with the lock set.
- **Shelf opening too tight:** an opening so low or so narrow that a gripper cannot enter level from the front carrying an item, or cannot reach the back of the working shelf.
- **Shelf will not hold an item:** a shelf that is not level, or so slick that a correctly stood item will not stay where it is put.
- **Unit not steady:** a pantry unit that rocks or walks when an item is drawn forward.
- **Zone label missing or unreadable at the start:** a label that has come off, is torn, or cannot be read from the front before recording starts.
- **Defective item:** one that leaks, that will not stand on its own base, whose lid or cap comes off in a correct body grip, or that slips out of a correct grip.
- **Zone has no room:** a stretch of shelf that will not hold its three items standing one behind the other with clear space between them.
- **Box will not collapse:** a box whose bottom flaps are taped or tucked so it will not fold flat under a correct push, or whose walls are too stiff to fold.
- **Box will not stand:** a box that folds or tips on its own before the shelving is done.
- **Stand will not stay put:** a stand or box that slides or tips when an item is lifted straight up out of it, or a stand too small for the flat box to lie on.
- **A place turns out to sit outside its arm's comfortable reach** with the base correctly parked, so a column of the box, a zone of the working shelf, or a corner of the box cannot be reached without extending or folding the arm, or without reaching into the other arm's lane.

Annotation subtasks (from SOP)

1. Slide the older item forward in one zone

2. Lift one item out of its column in the delivery box
3. Stand one new item down at the back of its zone
4. Stand one new item down between the older item and the first new item
5. Turn a held item until its label faces the front
6. Hold the back-left corner of the empty box
7. Push the front-right corner to collapse the box
8. Tip the collapsed box flat onto the stand
9. Return both arms home and end the episode